

GLA UNIVERSITY MATHURA

DEPARTMENT OF
COMPUTER ENGINEERING AND APPLICATIONS

LECTURE NOTES

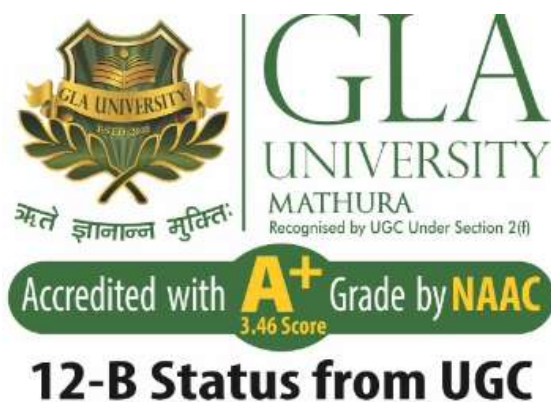
ON

BCAC 0019: Web Technology -1

**For
BCA – 1 Year**

Presented BY

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GLA UNIVERSITY MATHURA

BCAC 0815 : Web Technology

Objective: The objective is to provide a comprehensive study of the frontend. It stresses the strengths of Web, which provide students with the means of writing efficient, maintainable, and portable code.

Credits: 03

L-T-P-J: 3-0-0-0

Content	Teaching Hours
Introduction to Web Designing: Web Page, Website, Web Browser, WWW, Internet, e-Mail HTML : Introduction, Basic Elements, Attributes, Headings, Paragraphs, Styles, Formatting, Comments, Colors, Links, Images, Tables, Lists Overview of HTML: HTML Markup Tags, Linking, Tables and Form. Form Controls: Text Field, Password Field, Multiline Text Area, Drop Down List, Check Box, Radio Buttons, Scrolled List, Reset Button, Submit Button, File Field. : Introduction, Document Object Model (DOM)	22
Overview of DHTML: DHTML, Introduction, Document Object Model (DOM) CSS: Introduction to Cascading Style Sheets (CSS), Types of CSS, Selectors, Comments, Colors, Backgrounds, Borders, Rounded Corners, Border Images, Color Keywords, Margins, Padding, Height/Width, Text, Fonts, Links, Lists, Tables. Advanced CSS : Shadows Effects, Text Effects, Tooltips, Style Images	20

Text Books:

- Thomas Powell, "HTML & CSS: The Complete Reference, Fifth Edition, 2010, McGraw Hill Education India

Reference Books:

- Deborah S. Ray, Eric J. Ray, "Mastering HTML 4", 2nd Ed, Wiley India
- Steven M. Schafer, "HTML, XHTML, and CSS Bible", 5ed Wiley India
- John Duckett, "Beginning HTML, XHTML, CSS, and JavaScript", Wiley India
- Wendy Willard, HTML: A Beginner's Guide, Fifth Edition 2013, TMH

Outcome: After completion of course, student will be able to:

- CO1: Understand the concepts of www, evolution of www and internet and various client server architecture.
- CO2: Develop a web page with paragraphs, images, links, and lists and tables.
- CO3: Understand the concept of adding style in html document.
- CO4: Develop a web page using CSS

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO3/PSO2
CO2	PO3/PSO2
CO3	PO5/PSO2
CO4	PO4/PSO1

CONTENTS

Lecture 01:	overview of FIP
Lecture 02:	Pre Class Test
Lecture 03:	BASIC CONCEPT WEB PROGRAMMING
Lecture 04:	BENEFITS OF HTML WITH VERSIONS
Lecture 05:	Basics of HTML Programming Tools Like Notepad, Edit plus etc
Lecture 06:	Discussion on Web Page and Web Site
Lecture 07:	Basic of WWW, internet, http protocols and Email
Lecture 08:	HTML Elements and Attributes
Lecture 09:	Heading, Paragraph and style tags
Lecture 10:	Web Site Project framework analysis
Lecture 11:	Formatting methods, Colors settings
Lecture 12:	Comments, Link & Image Tags
Lecture 13:	Table Tags, Hyperlink, Marquee Tags
Lecture 14:	Table Animations
Lecture 15:	Image formatting
Lecture 16:	Form controls overview
Lecture 17:	Project prototype discussion
Lecture 18:	Text field, Password, Multiline Text Area Tags
Lecture 19:	Drop Down List and Checkbox
Lecture 20:	Radio Button, Check list, File Field Tags
Lecture 21:	Class Test and Mock Interview
Lecture 22 & 23:	Query Session and Project evaluation
Lecture 24:	Dom overview, comparative discussion on HTML & DHTML
Lecture 25:	Overview on CSS and its types
Lecture 26:	Selectors implementation in CSS with HTML
Lecture 27:	Colour Background, Border, Rounded Corner, Shadow effects on images
Lecture 28:	Border effects, Margin effects
Lecture 29:	Padding Concept & Font Family implementation
Lecture 30:	Table formatting
Lecture 31:	Evaluation [PI & Test]
Lecture 32:	Query Session and Project evaluation
Lecture 33:	Advance CSS overview
Lecture 34:	Shadow effects on Text & Image
Lecture 35:	Text Effects with CSS
Lecture 36:	Tool Tip implementation
Lecture 37:	Style Image
Lecture 38:	Dynamic Website Template Discussion
Lecture 39:	e portfolio guidelines
Lecture 40:	Introduction for advance Web Technologies like PHP

INTRODUCTION:

HTML stands for Hypertext Markup Language. It is a Language used to create Web Pages or Hypertext document. A Markup Language is a set of instructions often called TAGS which can be added to text files. HTML is only a formatting language is not a programming language. The idea behind hypertext is that instead of reading text in a right linear structure we can easily jump from one point to another point. HTML is all about specifying the structure and format of our webpage i.e, it is mainly used for describing the structure document.

HTML is platform independent i.e, for example if we can access internet, we can access WORLD WIDE WEB (WWW) irrespective of client OS and OS of the web server are accessing. So, we can view one download HTML files on www through browser.

Elements of a web document are labeled through the usage of HTML tags. It is the tags that describe the document. Anything that is not a tag will be displayed in the document itself. HTML does not describe any page layout i.e, for example, word for windows have different styles for headings, font size etc. But HTML doesn't have all these. Based on the Platforms, appearance of any element will change. The formatted text will appear differently on different machines / Platforms. By separating the Structure of the document and appearance, a Program that reads and Understands HTML can make formatting decision based on capabilities of individual Platform. Web Browsers are best examples of HTML formatters.

Advantages of HTML:-

- A HTML document is small and hence easy to send over the net. It is small because it doesn't include format information.
- HTML documents are cross platform compatible and device independent. We need a HTML readable browser to view them.

Basic HTML tags:-

(1) <!doctype> :

This tag formally starts an HTML document and it also indicates the version of HTML used.

<!doctype HTML PUBLIC “//w3c//DTDHTML Q.o//EN”>

(2) <HTML>:

Every HTML document starts with a <html> tag and it is always the first tag in a htmlpage and indicates that the document is a HTML document. The end tag <html> is </html>.

Example:

<html>..... </html>

(3) <head>:

It contains the head of an html document, which holds about the document such as title. Each property defined html page should have a head which we create with <head> tag. It has header information and it is displayed at the top of the browser. Each tag for

<head> is </head>.

<head></head>

(4) <title>:

It contains the title of the html document which includes the content that will actually appear in the web browser. The entire content of the web page is placed in the pages

<body> tag. The end tag <body> is </body>

<title> </title>

(5) <body>:

It contains the body of the HTML Document, which includes the content that will actual appear in the web browser. The entire content of the webpage will be placed in the pages <body> tag. The end tag of the

<body> tag will be </body>.

<body></body>

STRUCTURE OF THE HTML PROGRAM:-

The HTML Program is generally divided into two sections i.e. head and body. We use <head> and <body> tags to indicate these two sections. <head> section holds the header

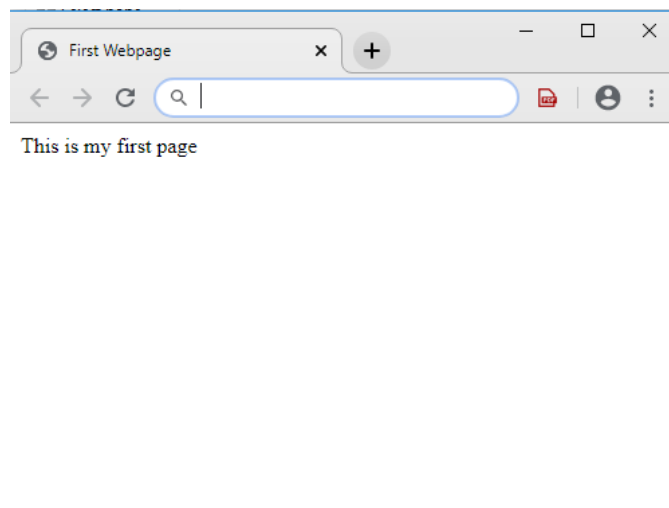
information of a webpage document indicated by a title that is provided by using <title> tag in the <head>. The title helps us to refer to the webpage. <Body> section contains the content which we want to display within the webpage. Anything that is not a tag will be displayed within the webpage.

Example:

```
<html>
  <head>
    <title>First Webpage</title>

  </head>
  <body>
    This is my first page
  </body>
</html>
```

Output:



Attribute:

An Attribute is a Keyword we use in an opening tag to give more information to the web browser. HTML tags tell the web browsers how to format and organize our WebPages. But we can customize tags using attributes. The Format of an attribute is:

<tag name Attribute=value>

Attributes of the <body> tag:

(1) Background:

The URL or a graphic file to be used in the filling the browser's Background.

(2) Bgcolor:

The color of the browser's background.

(3) Bg properties:

It Indicates if the background should scroll when text does. If we set it to "FIXED", the background will not scroll when the text does.

(4) Bottom margin:

Specifies the bottom margin, the empty space at the bottom of the document in pixels.

(5) Id:

It is a unique alphanumeric identifier for the tag which we can use to refer to it.

(6) Language:

Scripting language used for the tag.

(7) Left margin:

Specifies the left margin, the empty space at the left of the document.

(8) Margin height:

Gives the height of the margin at the top and bottom of the page in pixels.

(9) Margin Width:

Gives the width of the left and right margins of the page in pixels.

(10) Right margin:

It specifies the right margin, the empty space to the right margin of the document in pixels.

(11) Scroll:

It specifies whether a vertical scrollbar appears to the right of the document can be yes(or) no.

(12) Style:

Inline style indicating how to render the element.

(13) Text:

Color of the in the document.

(14) Top margin:

It specifies the top margin the space at the top of the document in pixels.

(15) Link:

It specifies the color of hyperlinks that have not yet been visited.

(16) Alink:

It specifies the color of hyperlinks as they are being clicked.

(17) Vlink:

It specifies the color of hyperlinks as they have been visited.

(18) <!--> Comment tag:

Annotates a web page with a comment. In the HTML that we can by looking at the HTMLbut it will not be displayed in the web browser.

<!-- This is a comment----- >

Formatting with HTML tags:

To set the actual style of text as displayed in a web page we can text style tags. There are a number of ways to apply styles to text.

(1) :

It creates a bold text i.e, sets the text style to bold.

Attributes:

a. Id:

It is a unique alphanumeric identifier for the tag which we can use to refer to it.

b. Style:

The Inline style indicating how to render the element.

Example:

```
<html>
```

```
  <head>
```

```
    <title>Using Bold Tag </title>
```

```
  </head>
```

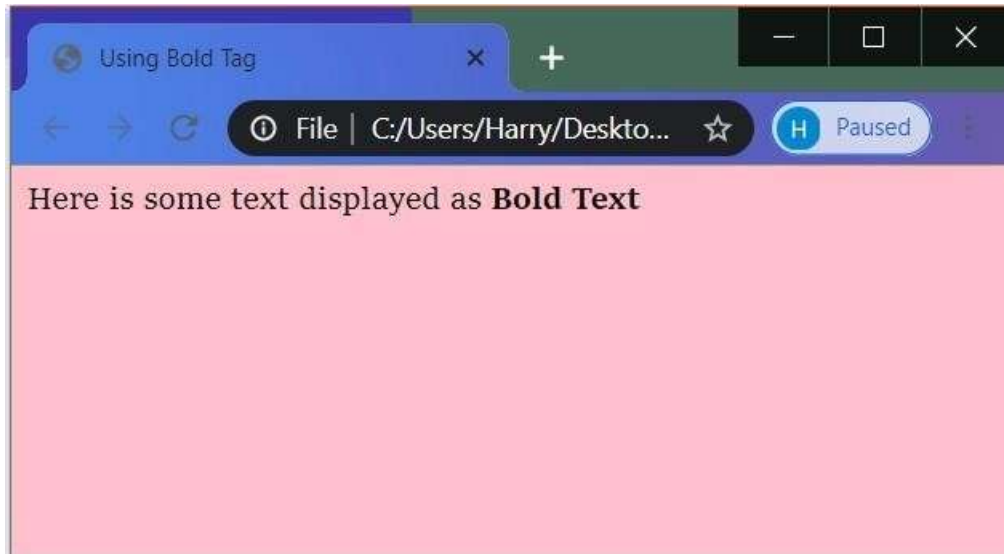
```
  <body bgcolor="pink">
```

```
    Here is some text displayed as <b> Bold Text </b>
```

```
</body>
```

```
</html>
```

Output:



(2)<I>:

It displays text in Italics.(3)

<U>:

It displays text in Underlined text.(4)

<P>:

It displays the Paragraph text.

Example:

```
<html>
```

```
<head>
```

```
<title> Using Styles </title>
```

```
</head>
```

```
<body bgcolor="pink">
```

```
<p> This is a paragraph <br>
```

```
Here is some text that is <i> Displayed in Italics </i>
```

```
<br>Here is some <u> Underlined text </u>
```

```
</body>
```

```
</html>
```

Output:



(5) <s> and <strike>:

It Displays text in strike through style. The <s> and <strike> tags are used for the same effect. HTML 2 used <strike>, HTML 3 called it <s>, HTML 3.2 caused it <strike> again.

(6) <big>:

Renders text in a bigger font than the current default.

(7) <small>:

Renders text in a smaller font than the current default.

Example:

```
<html>
```

```
  <head>
```

```
    <title> Using Big and small tags </title>
```

```
  </head>
```

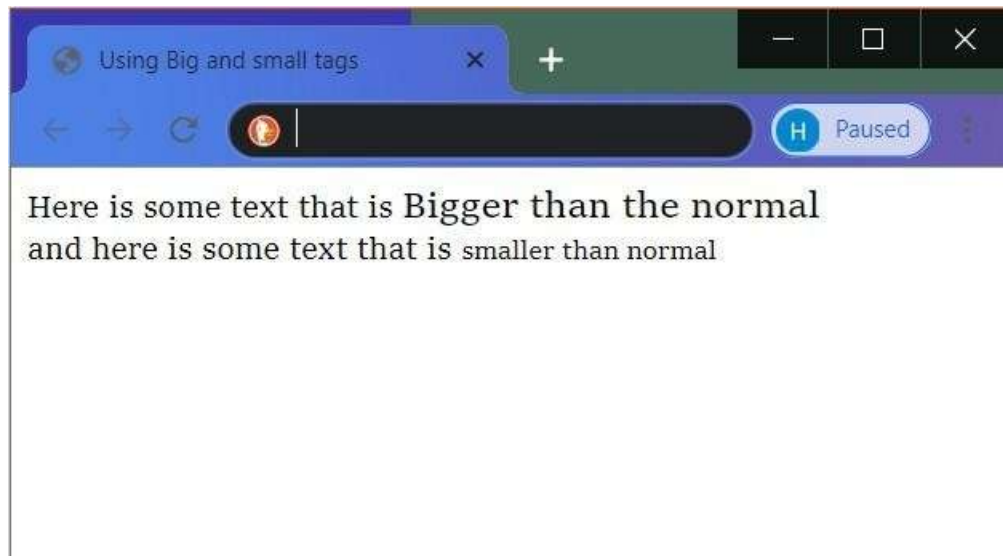
```
  <body>
```

```
    Here is some text that is <big> Bigger than the normal </big> <br>and here is  
    some text that is <small> smaller than normal </small>
```

```
  </body>
```

```
</html>
```

Output:



(8) `<sub>`: It Styles the text as a subscript.

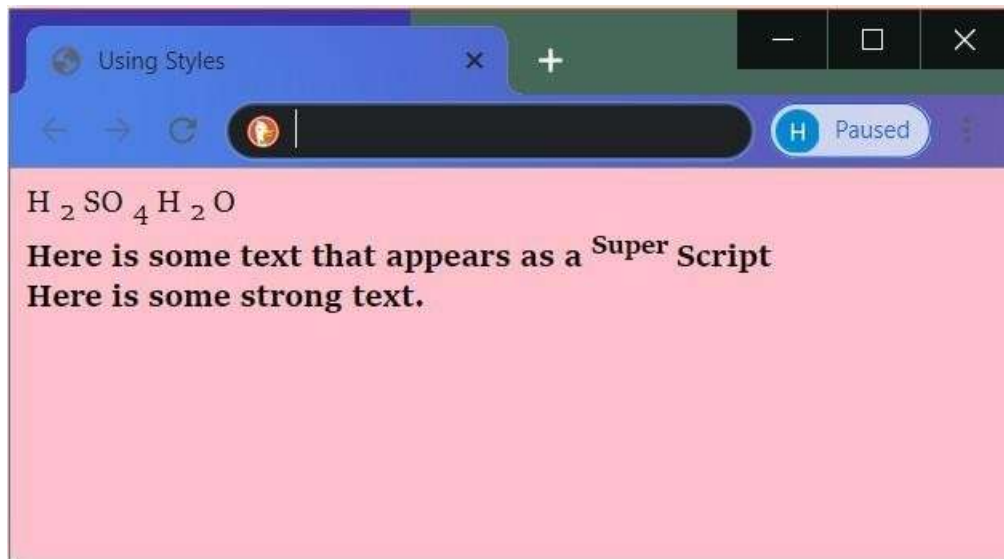
(9) `<sup>`: It Styles the text as a superscript.

(10) `<strong>`: Emphasizes text strongly, usually rendered in bold.

Example:

```
<html>
  <head>
    <title>Using Styles </title>
  </head>
  <body bgcolor="pink">
    H <sub> 2 </sub> SO <sub> 4 </sub> H
    <sub> 2 </sub> O
    <br>
    <b>Here is some text that appears as a <sup> Super </sup> Script <br>Here is some
    <strong> strong </strong> text.
  </body>
</html>
```

Output:



(11)Headings:

<h1>,<h2>,<h3>,<h4>,<h5> & <h6>

The heading element tags are <h1>,<h2>,<h3>,<h4>,<h5>,<h6>. These elements create the headings in our web pages by displaying bold text in a variety of sizes <h1> being larger <h6> being smaller.

Example:

```
<html>
  <head>
    <title> Heading tags </title>
  </head>
  <body bgcolor="pink">
    <center>
      <h1> Using Heading Tags</h1><br>
      <h1> RGM CET </h1><br>
      <h2> RGM CET </h2><br>
      <h3> RGM CET </h3><br>
      <h4> RGM CET </h4><br>
      <h5> RGM CET </h5><br>
```

```
<h6> RGM CET </h6><br>  
</center>  
</body>  
</html>
```

Output



(12):

This tag will give us an option to select text size, color and face.

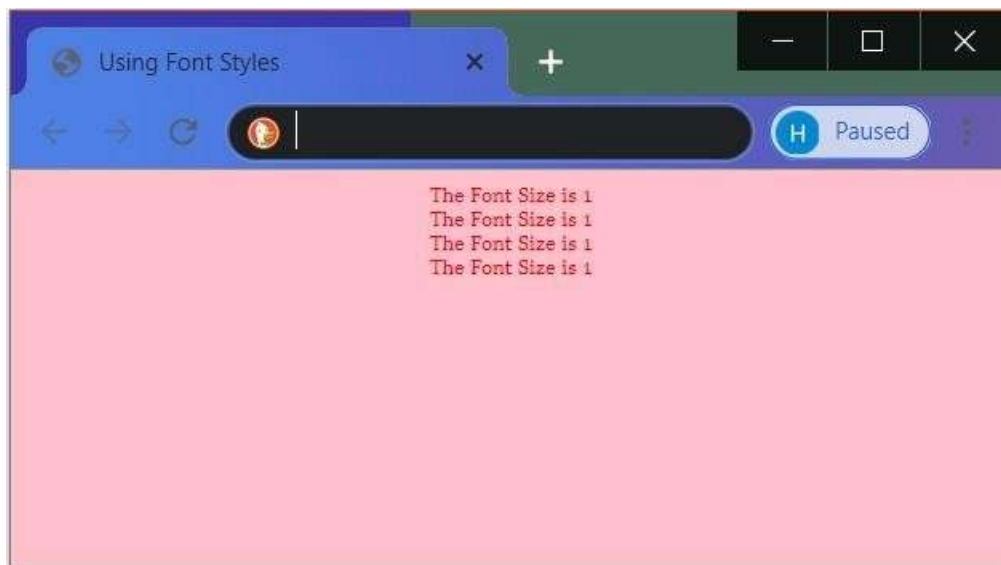
Attributes:

- color: Color of the text.
- Size: Size of the text in points
- Face: The font face can be a list of names separated by commas.
- Id: Unique alphanumeric identifier for a tag, which we can use to refer to it.

Example:

```
<html>
  <head>
    <title> Using Font Styles </title>
  </head>
  <body bgcolor="pink">
    <center>
      <font size="1" color="red"> The Font Size is 1 </font> <br>
      <font size="10" color="yellow"> The Font Size is 10 </font> <br>
      <font size="20" color="orange"> The Font Size is 20</font> <br>
      <font size="30" color="aqua"> The Font Size is 30 </font> <br>
    </center>
  </body>
</html>
```

Output



(13) <marquee> tag:

Displays scrolling text in a marquee style.

Attributes:

a. Align:

Sets the alignment of the text relative to marquee. Set to:

Top(default), middle (or) bottom.

b. behavior:

Sets how the text in the marquee should move can be scroll (default), slide(text enters from one side and stops at the other side), alternate (text seems to bounce from one side to the other).

c. bgcolor:

It sets the background color for the marquee box.

d. Direction:

Sets the direction the text should scroll can be left, right, down or up.

e. Height:

It specifies the height of the marquee.

f. Loop:

Sets how many times we want the marquee to cycle. Is set to positive integer or -1 for continuous cycling.

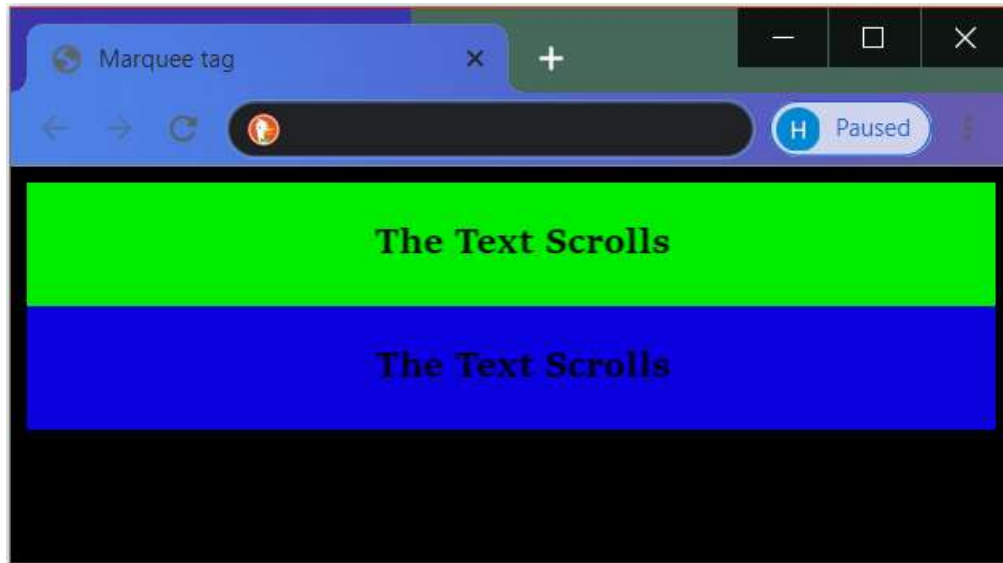
g. ScrollDelay:

Sets the number of the milliseconds between each successive display text.

Example:

```
<html>
  <head>
    <title> Marquee tag </title>
  </head>
  <body bgcolor="pink">
    <marquee align="top" loop="infinite" behavior="scroll" bgcolor="red"
direction="right"> <h3> The Text Scrolls </h3></marquee>
    <marquee align="middle" loop="infinite" behavior="slide" bgcolor="blue"
direction="left"> <h3> The Text Scrolls </h3></marquee>
  </body>
</html>
```

Output



(14) `<pre>` tag(preformatted text):

`<pre>` marks the text as preformatted text i.e, all the spaces and carriage returns as rendered exactly as you type them.

Example:

```
<html>
```

```
  <head>
```

```
    <title> Pre Tag </title>
```

```
  </head>
```

```
  <body bgcolor="pink">
```

```
    <center>
```

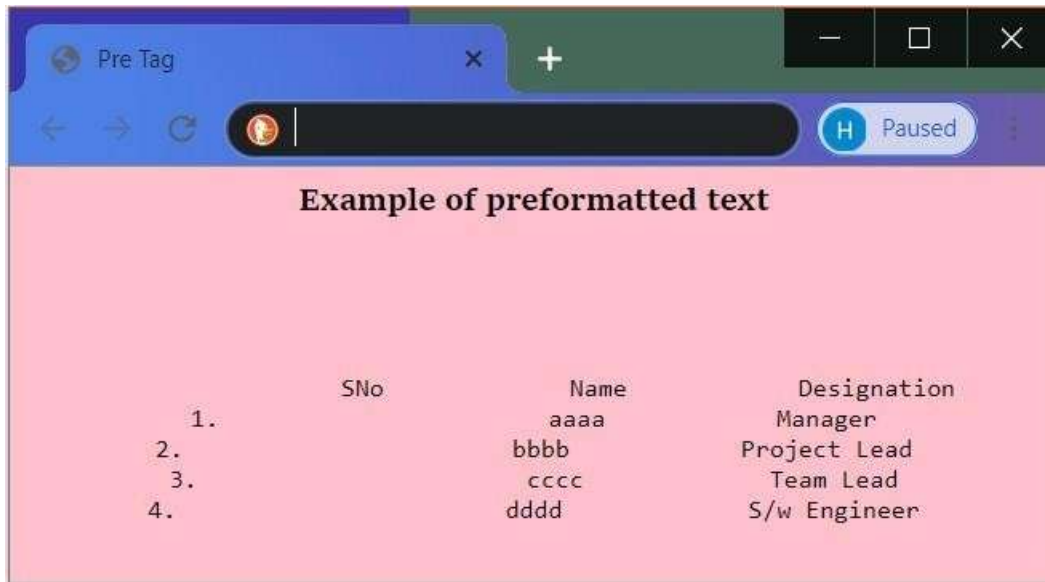
```
    <h4> Example of preformatted text </h4> <br> <br>
```

```
    <pre>
```

```
      SNo      Name      Designation
      1.      aaaa      Manager
      2.      bbbb      Project Lead
      3.      cccc      Team Lead
      4.      dddd      S/w Engineer
    </pre>
```

```
</center>
</body>
</html>
```

Output:



Lists:

Lists let us display information in a compact, right format. There are three kinds of lists:

1. Unordered List
2. Ordered List
3. Definition List

Unordered List:

An Unordered list is a list of items that are marked with a bullet. The Unordered list is created by using `<ul>` tag and the list items in the list are created by `<li>` tag and the list items in the list are created by `<li>` tag.

```
<ul>
  <li>List Item 1 </li>
  <li>List Item 2 </li>
</ul>
```

Example:

```
<html>
```

```
<head>
  <title> Creating Unorder List </title>
</head>
<body bgcolor="pink">
  <h1 align="center"> Creating Unorder List</h1>
  <h1 align="center">List of Colleges in Kurnool</h1>
  <ul>
    <li>GPREC</li>
    <li>RGM CET</li>
    <li>GPCET</li>
  </ul>
</body>
</html>
```

Output



Creating Customized Unordered Lists:

We customized unordered lists by setting the “Type” attribute to three different values. DISC (default), SQUARE and CIRCLE which sets the type of bullet that appears before the list item.

Example:

```
<html>
<head>
  <title> Creating Unorder List </title>
</head>
<body bgcolor="pink">
  <h1 align="center"> Creating Unorder List</h1>
  <h1 align="center">List of Colleges in Kurnool</h1>
  <ul type="square">
    <li>GPREC</li>
    <li>RGM CET</li>
    <li>GPCET</li>
  </ul>
</body>
</html>
```

Output



Ordered List:

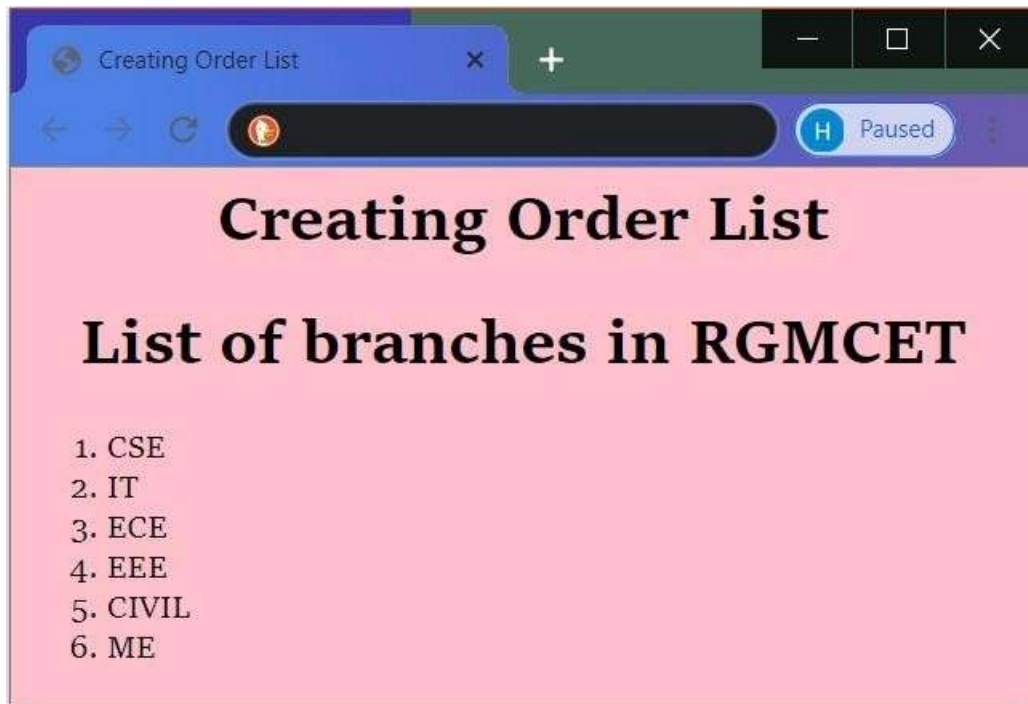
While the unordered lists display simple bullet before each list item. Ordered lists use a number system / lettering scheme to indicate that the items are ordered in some ways, ordered lists are

created by tag and the list items are created using tag.

Example:

```
<html>
<head>
    <title> Creating Order List </title>
</head>
<body bgcolor="pink">
    <h1 align="center"> Creating Order List</h1>
    <h1 align="center">List of branches in RGM CET</h1>
    <ol>
        <li>CSE</li>
        <li>IT</li>
        <li>ECE</li>
        <li>EEE</li>
        <li>CIVIL</li>
        <li>ME</li>
    </ol>
</body>
</html>
```

Output



Creating Customized Ordered Lists:-

We can customize the numbering system used in ordered lists by using the TYPE attribute, which we can set to these values:

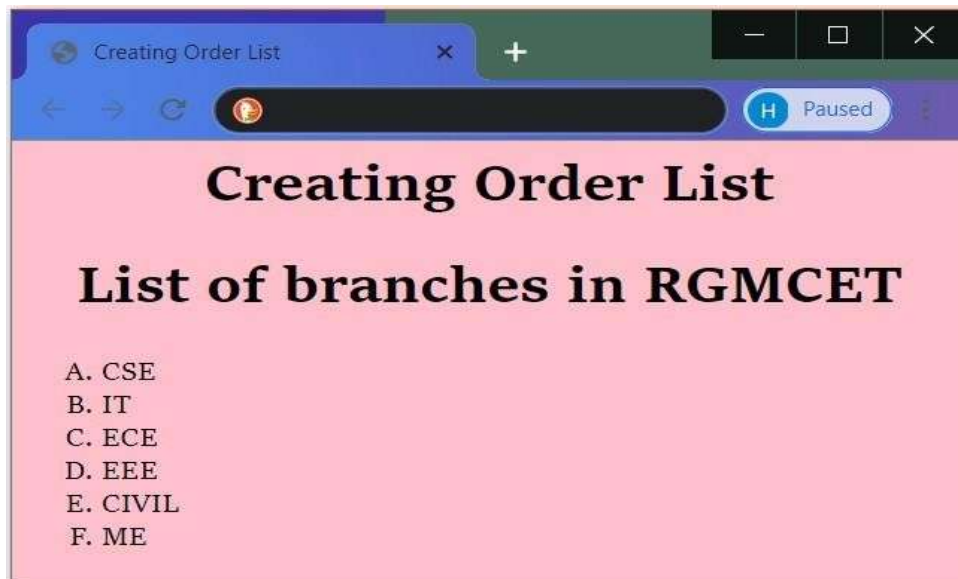
1. Default numbering system (1, 2, 3,)
- A. Uppercase Letters (A, B, C,
- a. Lowercase Letters (a, b, c, ...)
- I. Large Roman Numerals (I, II, III,)
- i. Small Roman Numerals (i, ii, iii,

Example:

```
<html>
<head>
<title> Creating Order List </title>
</head>
<body bgcolor="pink">
  <h1 align="center"> Creating Order List</h1>
  <h1 align="center">List of branches in RGM CET</h1>
```

```
<ol type="A">
  <li>CSE</li>
  <li>IT</li>
  <li>ECE</li>
  <li>EEE</li>
  <li>CIVIL</li>
  <li>ME</li>
</ol>
</body>
</html>
```

Output



Definition List:-

These lists include both definition terms as well as their definition. To create the definition lists we use `<dl>` tag. For creating definition terms we use `<dt>` tag and for data definitions we use

`<dd>` tag.

Example:

```
<html>
  <head>
```

```
<title>Creating Definition List</title>
</head>
<body bgcolor="pink">
  <h1 align="center">Definition List</h1>
  <dl>
    <dt>CSE<dd>Computer Science & Engineering
    <dt>ECE<dd>Electronics & Communication Engineering
    <dt>IT<dd>Information Technology
    <dt>EEE<dd>Electrical & Electronics Engineering
    <dt>CE<dd>Civil Engineering
  </dl>
</body>
</html>
```

Output



Nesting Lists:-

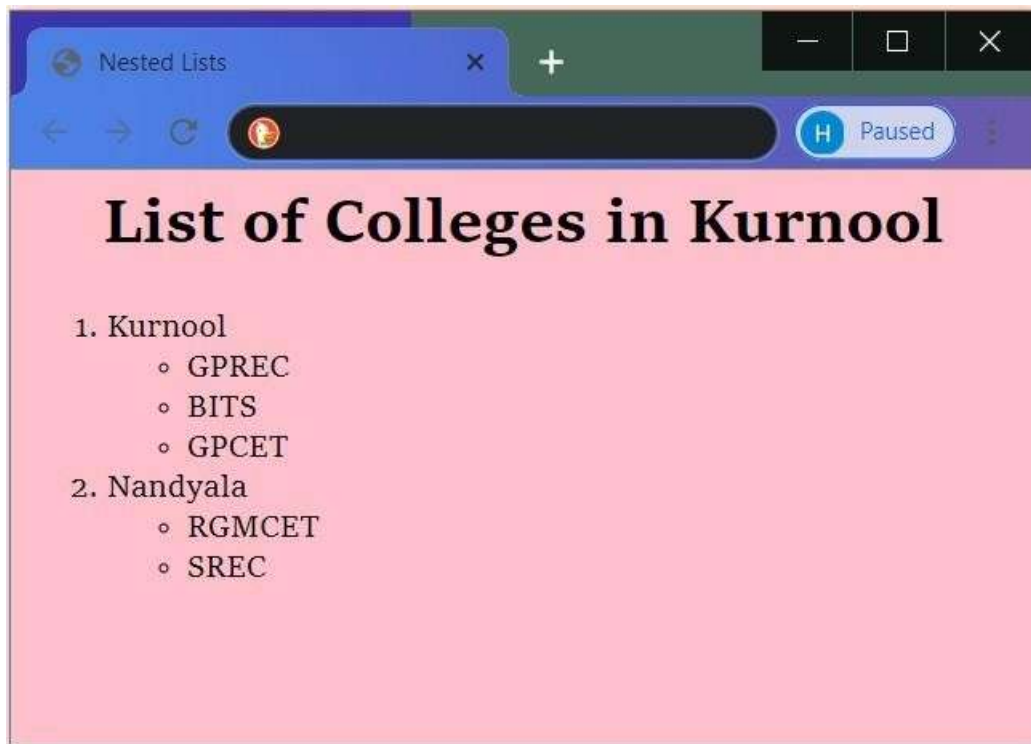
We have the capability of nesting lists inside other lists.

Example:

```
<html>
  <head>
    <title>Nested Lists</title>
  </head>
  <body bgcolor="pink">
    <h1 align="center">List of Colleges in Kurnool</h1>
  <ol>
    <li>Kurnool</li>
    <ul>
      <li>GPREC</li>
      <li>BITS</li>
    </ul>
  </ol>
```

```
<li>GPCET</li>
</ul>
<li>Nandyala</li>
<ul>
  <li>RGM CET</li>
  <li>SREC</li>
</ul>
</ol>
</body>
</html>
```

Output



Creating Hyperlinks:

What makes the web so effective is the ability to define links from one page to another. In web terms, a “hyperlinks” is a reference on the web. Hyperlinks can point to any resources on the web. An anchor is a term used to define a hyperlink destination inside a document. Format of anchor tag is:

```
<a href="address"> Line Text </a>
```

The <a> anchor tag has the following attributes.

1. href: It holds the target URL of the hyperlink.
2. Id: A unique alphanumeric identifier for the tag, which we can use to refer to it.
3. name: It specifies an anchor name, the name we want to use when referring to enclose items.
4. Target: This attribute defines where the linked document will be opened.

Example:

```
<html>
<head>
  <title>Creating Hyper Links</title>
</head>
<body bgcolor="pink">
  <center><h1>This is page 1</h1>
  <a href="page2.html">Click here</a>to goto page2
</center>
</body>
</html>
```

Output



Setting hyperlink colors:

The default color of hyperlinks in a page is blue. Hyperlink that we have already visited is displayed in violet and when we click a hyperlink, it turns red when the mouse button is down. We can set these colors in <body> tag attributes link, vlink(visited link), alink(active link).

Example

```
<html>
<head>
  <title>Setting Hyperlink colors</title>
</head>
<body bgcolor="pink" link="green" vlink="blue" alink="red">
  <center><h1>Setting Hyperlink colors</h1>
  <a href="login.html">Click here</a> to goto login page
</body>
</html>
```

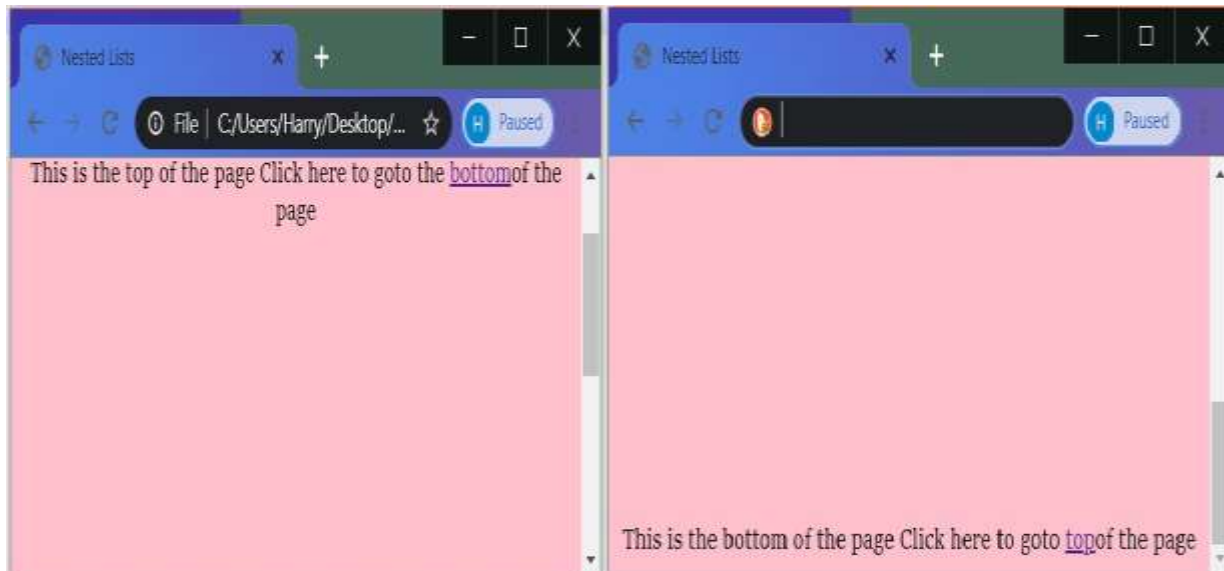
Output



Providing navigation with in the page:

```
<html>
  <head>
    <title>Nested Lists</title>
  </head>
  <body bgcolor="pink">
    <center><h1>Linking to a section in a page</h1>
    <a name="top">This is the top of the page</a>
    Click here to goto the <a target="#bottom">bottom</a>of the page
    <br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br>
    <br><br><br><br><br>
    <a name="bottom">This is the bottom of the page</a> Click here to
    goto <a target="#top">top</a>of the page
  </center>
</body>
</html>
```

Output



Creating HML tables:

A HTML table arranges data/information in terms of rows and columns. Tables are defined in HTML using `<table>` tag. A table is divided into rows and each row each divided into data cells(columns).The rows of table are created using `<tr>` tag and data cells are created by `<td>` tag.

`<tr>` - Table row

`<td>` - Table data

Format

```
<table>
  <tr>
    <td>row1,col1</td>
    <td>row1,col2</td>
  </tr>
  <tr>
    <td>row2,col1</td>
    <td>row2,col2</td>
  </tr>
</table>
```

- Heading in a table are defined with `<th>` tag

Format

```
<table>
  <tr>
    <th>heading 1</th>
    <th>heading 2</th>
  </tr>
  <tr>
    <td>data1</td>
    <td>data2</td>
  </tr>
</table>
```

Attributes of `<table>` tag:

- align : specifies the horizontal alignment of the table in the browser window, set to "left,center,right".
- background : specifies the URL of a background image to be used as backgroundfor the table.
- bgcolor : sets the background color of the table cells.
- border : sets the border width.
- bordercolor : sets the external border color of the entire table.
- cellpadding : sets the spacing between cell walls and content.
- cellspacing : sets the spacing between table cells.
- height : sets the height of the whole table.
- width : sets the width of the table.

Attributes of <tr> tag:

- align : specifies the horizontal alignment content in the table cells set to "left,center,right".
- bgcolor : sets the background color of the table cells.
- bordercolor : sets the external border color of the entire table.
- Valign : sets the vertical alignment of data, set to top, middle, bottom.

Alignment of <td> tag

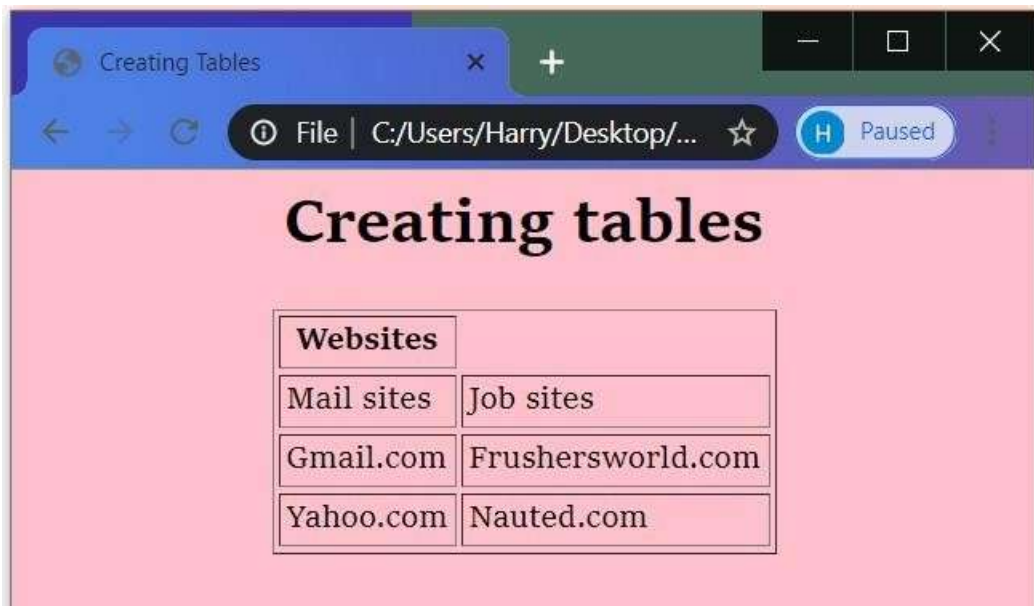
- align : specifies the horizontal alignment content in the table cells set to "left,center,right".
- bgcolor : sets the background color of the table cells.
- bordercolor : sets the external border color of the entire table.
- colspan : indicates the how many cell columns of the table this cell should span.
- rowspan : indicates the how many cell rows of the table this cell should span.

Example

```
<html>  
<head>
```

```
<title>Creating Tables</title>
</head>
<body bgcolor="pink">
<center><h1>Creating tables</h1>
<table border="1" cellpadding="3" cellspacing="3">
  <tr>
    <th colspan="2">Websites</th>
  </tr>
  <tr>
    <td>Mail sites</td>
    <td>Job sites</td>
  </tr>
  <tr>
    <td>Gmail.com</td>
    <td>Frushersworld.com</td>
  </tr>
  <tr>
    <td>Yahoo.com</td>
    <td>Nauted.com</td>
  </tr>
</center>
</table> </body></html>
```

Output



Advanced Table elements :

- `<caption>` : the element is an optional element and it is used to provide a string which describes the content of the table, it must follow the table element.
- `<thead>` : The rows in a table can be graduated one or more times we can create a table by using this `<thead>`.
- `<tbody>` : creates a table body when grouping rows.
- `<tfoot>` : Creates a table foot when grouped in rows.

Example:

```
<html>
<head>
  <title>Advance Table Elements</title>
</head>
<body bgcolor="pink">
  <h1 align="center">Contents of Web Technologies</h1>
  <center>
    <table border="2">
```

```
<caption>Subject Description</caption>
<thead>
  <tr> <td colspan="2">Advance Java Programming</td>
</thead>
<tbody>
  <tr> <td>Units</td>
    <td>Contents</td>
  </tr>
  <tr> <td>I</td>
    <td>HTML & CSS</td>
  </tr>
  <tr> <td>II</td>
    <td>JavaScript</td>
  </tr>
  <tr> <td>III</td>
    <td>XML</td>
  </tr>
</tbody>
<tfoot align="center">
  <tr>
    <td colspan="2">The table foot</td>
  </tr>
</tfoot>
</table>
</center>
</body>
</html>
```

Output



Nesting of Tables:

```
<html>
<head>
  <title>Nesting of Tables</title>
</head>
<body bgcolor="pink">
  <center><h1>Nested tables</h1>
  <table border="1" cellpadding="3" cellspacing="3">
    <tr>
      <td>
        <table border="2">
          <tr>
            <th>Mail sites</th>
            <th>Job sites</th>
          </tr>
```

```
<tr>
    <td>Gmail.com</td>
    <td>Frushersworld.com</td>
</tr>
<tr>
    <td>Yahoo.com</td>
    <td>Nauted.com</td>
</tr>
</table>
</td>
<td>
    <table border="2">
        <tr>
            <th>Number</th>
            <th>Words</th>
        </tr>
        <tr>
            <th>1</th>
            <th>One</th>
        </tr>
        <tr>
            <th>2</th>
            <th>Two</th>
        </tr>
    </table>
</td>
</tr>
</table>
</center>
</body>
```

</html>

Output



Images in HTML:

In HTML we have the capability of displaying images in a webpage. These images must be in a format that the web browser can handle, such as Graphics Interchange Format (GIF), Joint Photographic Experts Group (JPEG), and for some browsers Portable Network Graphics (PNG) formats.

Displaying images in a webpage is done by using tag

Format

```

```

Attributes of tag:

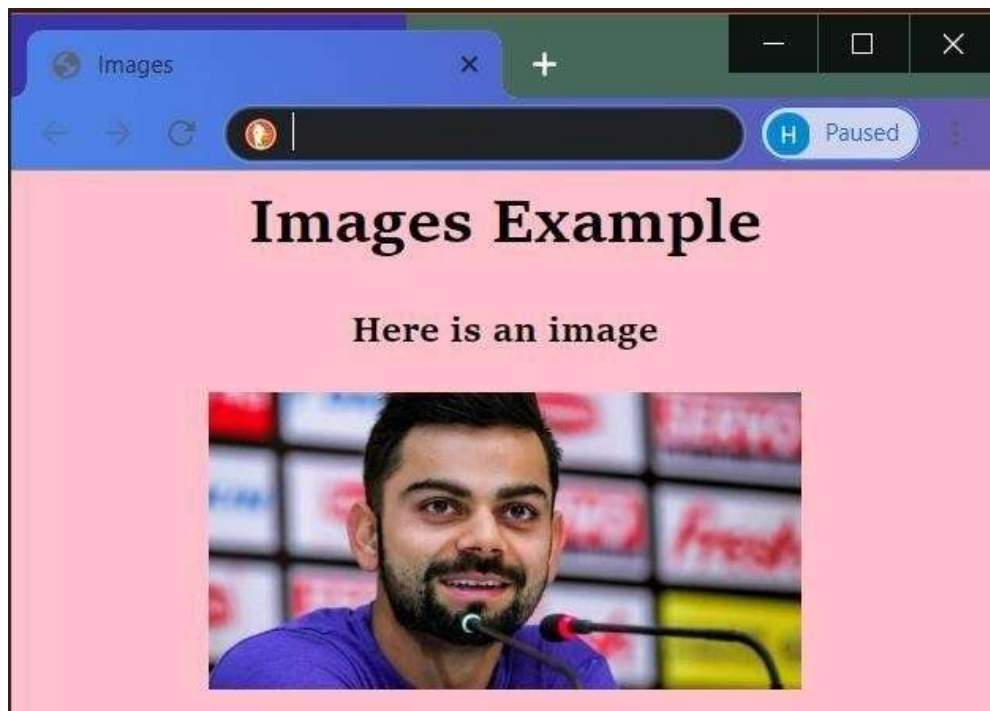
- alt : this attribute is used to specify text to be displayed in place of image for browser that cannot handle graphics.
- src : specifies the URL of the image to display.
- border : sets the border for the image.
- height : indicates the height of the image.
- width : indicates the width of the image.

- `hspace` : sets the horizontal space around the image.
- `vspace` : sets the vertical space around the image.

Example

```
<html>
<head>
  <title>Images</title>
</head>
<body bgcolor="pink">
  <center>
    <h1>Images Example</h1>
    <h3>Here is an image</h3>
    
  </center>
</body>
</html>
```

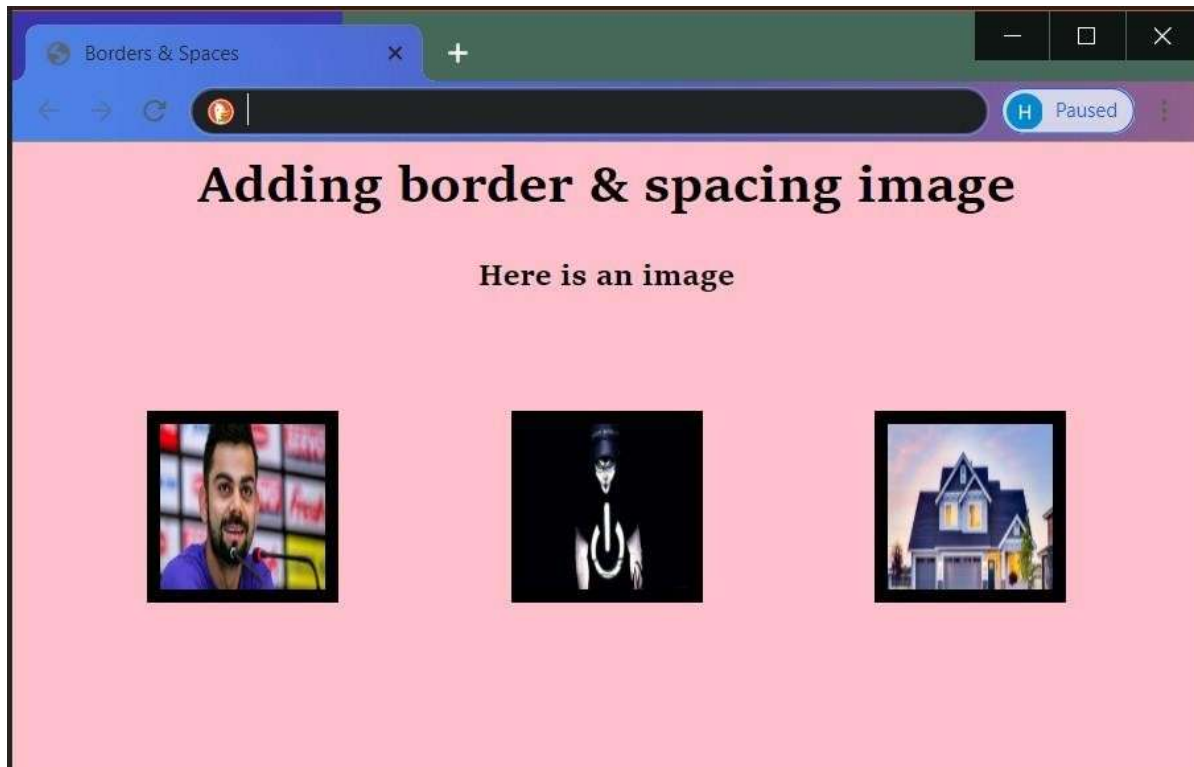
Output



Adding borders and spaces around image:

```
<html>
<head>
  <title>Borders & Spaces</title>
</head>
<body bgcolor="pink">
  <center>
    <h1>Adding border & spacing image</h1>
    <h3>Here is an image</h3>
    
    
    
  </center>
</body>
</html>
```

Output:



Creating HTML Forms:

Form is a collection of various HTML control files , buttons ,checkboxes , radio buttons ,text fields et., and they use to send the data to the server. There are several form elements.

- Button : `<input type="button">` :- are the standard clickable buttons.
- Checkbox : `<input type="checkbox">` :- displayed usually as a small box with a checkmark in it. The use can toggle the checkbox on or off by checking the checkbox
- Customizable Buttons : `<button>` :- display images one other HTML inside itself.
- File uploading controls : `<input type="file">` :- allow the user to upload files to the server.
- Hidden controls : store data that is not visible to users unless they view the web pagesource code.
- Image controls : `<input type="image">` :- are like submit buttons except that they are images the user can click.
- Password controls : `<input type="password">` :- are like text fields , but each typed character displaying by an asterisk or instead any character.

- Radio buttons : `<input type="radio">` :- displaying usually as a circle which when selected displayed a dot in the middle. These controls are much like checkboxes except that they work in to mutually exclusive at a time.
- Reset button : `<input type="reset">` : - allow the user to clear all the data they has entered. When the user click reset button all controls in the form are removed to that original state displaying the data they had when they first appeared.
- Selection : Works much like drop down list boxes also called select controls Format is:

```
<select>
    <option>Item1</option>
    <option>Item1</option>
    <option>Item1</option>
</select>
```

- Submit button : when we click the button all the data in the form will be sent to web serverfor processing.
- Text area : are two dimensional text fields allowing user to enter more than one line of text.
Format is: `<text area>`
- Text fields : allow the user to enter one line of text also called a textbox Format is :
`<input type="text">`

In order to create form we use `<form>` tag

Formatis :

```
<form>
|
|
</form>
```

Attributes of `<form>` tag:

- name : gives the name of the form so that we can return it in code . Set to an alphanumeric string.
- target : indicates a named frame for browser to display the form results.

- method : indicates a method or protocol for sending data to the target action URL.
- action : gives the URL that that will handle the form data.

Example

Registration.html

```
<html>
<head>
<title>HTML Form</title>
</head>
<body bgcolor="pink">
<center>
<form name="form1">
<table border="0" cellpadding="4" cellspacing="4">
<caption>Registration form</caption>
<tr>
<th>Name</th>
<td><input type="text" name="name" /></td>
</tr>
<tr>
<th>Password</th>
<td><input type="password"/></td>
</tr>
<tr>
<th>Enter your address</th>
<td><textarea rows="5" cols="10"></textarea></td>
</tr>
<tr>
<th>Enter your email</th>
<td><input type="email"/></td>
</tr>
</tr>
```

```
<th>Enter your mobile</th>
<td><input type="number"/></td>
</tr>
<tr>
<th>Select your gender</th>
<td>
male<input type="radio" name="g" value="m"/>
female<input type="radio" name="g" value="f"/>
</td>
</tr>
<tr>
<th>Language prference</th>
<td>
English<input type="checkbox" value="" />
Telugu<input type="checkbox" value="" />
Hindi<input type="checkbox" value="" />
</td>
</tr>
<tr>
<th>Select your DOB</th>
<td><input type="date"/></td>
</tr>
<tr>
<td><input type="submit" value="Register"/></td>
<td><input type="reset" value="Cancel"/></td>
</td>
</tr>
</table>
</form>
</body>
```

</html>

Output



The screenshot shows a web browser window with a single tab titled 'HTML Form'. The address bar contains the text 'Search DuckDuckGo or type a URL'. The main content area displays a registration form on a pink background. The form is titled 'Registration form' and contains the following fields and controls:

- Name**: A text input field.
- Password**: A text input field.
- Enter your address**: A text area.
- Enter your email**: A text input field.
- Enter your mobile**: A text input field.
- Select your gender**: Two radio buttons labeled 'male' and 'female'.
- Language preference**: Three checkboxes labeled 'English', 'Telugu', and 'Hindi'.
- Select your DOB**: A text input field with the placeholder 'dd-mm-yyyy'.
- Register**: A button.
- Cancel**: A button.

Working with Frames:

HTML frames allow user to present documents in multiple views which may be independent windows or sub windows. To divide a webpage into multiple parts and load different pages in a single web page we use the concept of frames. To do this we use “<frameset>” tag which indicates the browser that the webpage window has a frame. We can divide it into rows and columns by using attributes such as ‘rows’ & ‘cols’. In order to provide definition or each frame we use “<frame>” tag.

Format is :

```
<frameset rows="30%,70%">  
  <frame src="source page URL" name="frame name">  
  <frame src="source page URL" name="frame name">  
</frameset>
```

- <frameset> element actually takes place of <body> tag.

Attributes of <frameset> tag:

- border : used in the outermost <frameset> tag to set the border thickness for frames.
- border color : set the color of the borders for all frames in the frameset.
- frame border : set whether or not border for all frames in the frameset. Can be set to 'yes' or 'no' or '1' or '0'.
- frame spacing : set the pixel spacing between frames. set to the positive integers.
- cols : set the number of columns in the frameset. Separate the values assigned to this attribute with comma(,) each value represents width of column. Can be set to pixel values, percentages.
- Rows : set the number of rows in the frameset. Separate the values assigned to this attribute with comma(,) each value represents width of a column. Can be set to pixel values, percentages.

Attributes of <frame> tag:

- border color : set the color used for the frame border. This setting overrides the colors specified in the surrounding <frameset> element.
- frame border : sets whether or not border surround the frame. Can be set to 'yes' or 'no' or '1' or '0'.
- name : sets the name of the frame we can use named frames as target for <a>tag.
- scrolling : determines scrolling possible values are : auto, yes or no.
- src : specifies the URL of the frame document. If we don't specify a URL the frame will appear blank.

Creating vertical frames:

In order to display vertical frame we have use of 'cols' attribute.

Example

```
<html>
<head>
<title>Vertical Frames</title>
</head>
```

```
<frameset cols="30%,70%">
    <frame src=frame1.html>
    <frame src=frame2.html>
</frameset>
</html>
```

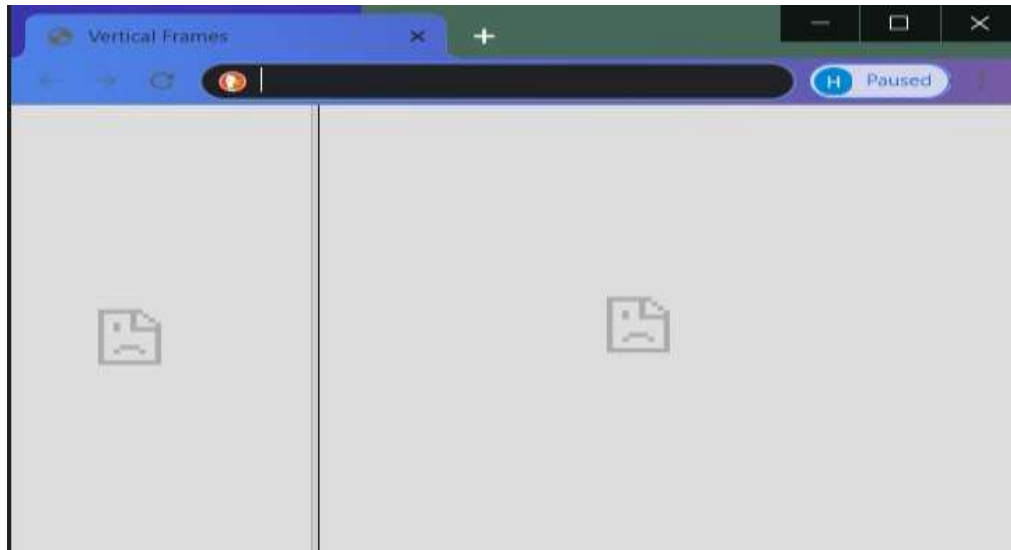
Frame1.html

```
<html>
<head>
    <title>page1</title>
</head>
<body>
    <h1>Web Technologies</h1>
</body>
</html>
```

Frame2.html

```
<html>
<head>
    <title>page2</title>
</head>
<body>
    <h1>Web Technologies</h1>
</body>
</html>
```

Output



Creating horizontal frames:

In order to display horizontal frame we have use of 'cols' attribute.

Example

```
<html>
<head>
    <title>Horizontal Frames</title>
</head>
<frameset cols="50%,50%">
    <frame src=page1.html>
```

```
<frame src=page2.html>
</frameset>
</html>

page1.html
<html>
<head>
    <title>page1</title>
</head>
<body>
    <h1 align="center">This is page1</h1>
</body>
</html>
```

```
page2.html
<html>
<head>
    <title>page2</title>
</head>
<body>
    <h1 align="center">This is page2</h1>
</body>
</html>
```

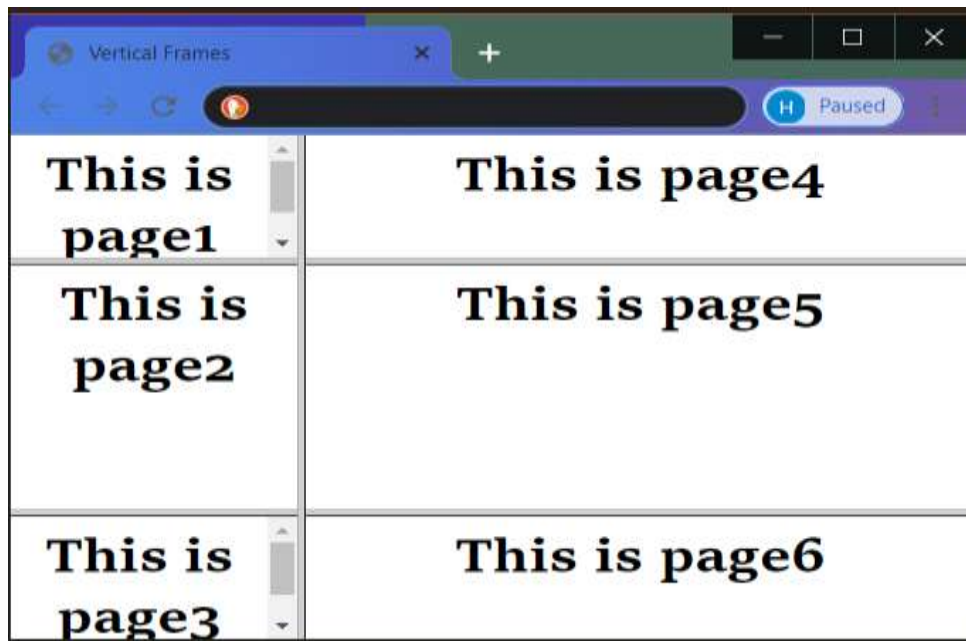
Output



Creating horizontal & vertical frames :

```
<html>
<head>
    <title>Horizontal Frames</title>
</head>
<frameset cols="30%,70%">
    <frameset cols="25%,50%,25%">
        <frame src=page1.html>
        <frame src=page2.html>
        <frame src=page3.html>
    </frameset>
    <frameset cols="25%,50%,25%">
        <frame src=frame1.html>
        <frame src=frame2.html>
        <frame src=frame3.html>
    </frameset>
</frameset> </html>
```

Output



<noframes> tag:

When the browser does not support frameset use `<noframes>` element to indicate to users that the browser does not support frames. The `<noframes>` element is ignored that handles frames.

Example

```
<html>
```

```
<head>
```

```
    <title>Vertical Frames</title>
```

```
</head>
```

```
<frameset cols="30%,70%">
```

```
    <noframes>Your browser does not support frames...</noframes>
```

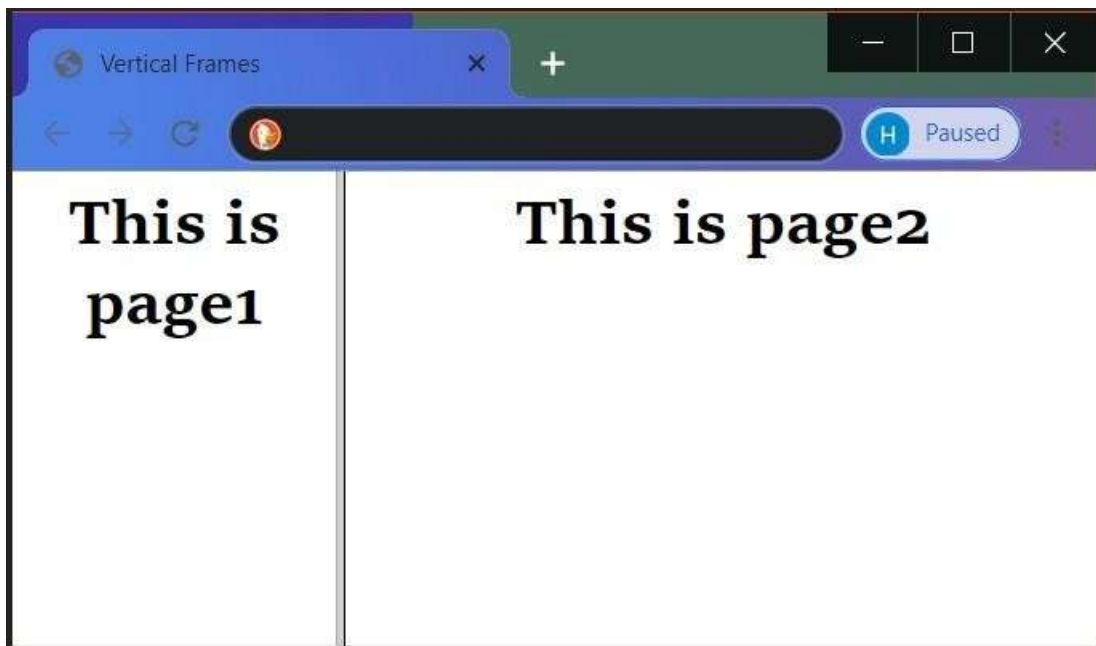
```
    <frame src=on.html>
```

```
    <frame src=two.html>
```

```
</frameset>
```

</html>

Output



Named frames:

One important aspect of working with frames is using named frames. When we give frame name , we can use as a target to load new page into the frame.

Example

<html>

<head>

 <title>Vertical Frames</title>

</head>

<frameset cols="40%,60%">

 <frame src=menu.html>

 <frame src=default.html name="display">

</frameset>

</html>

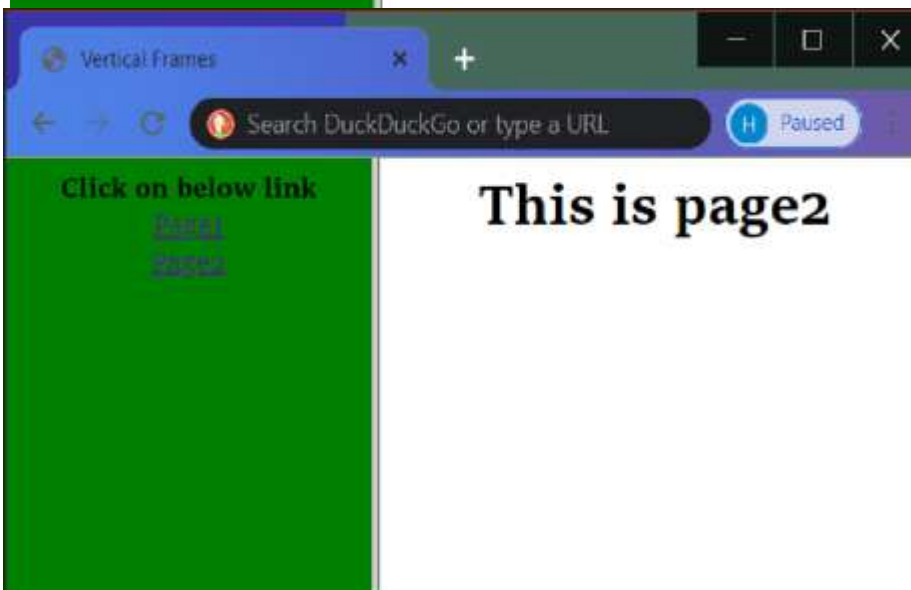
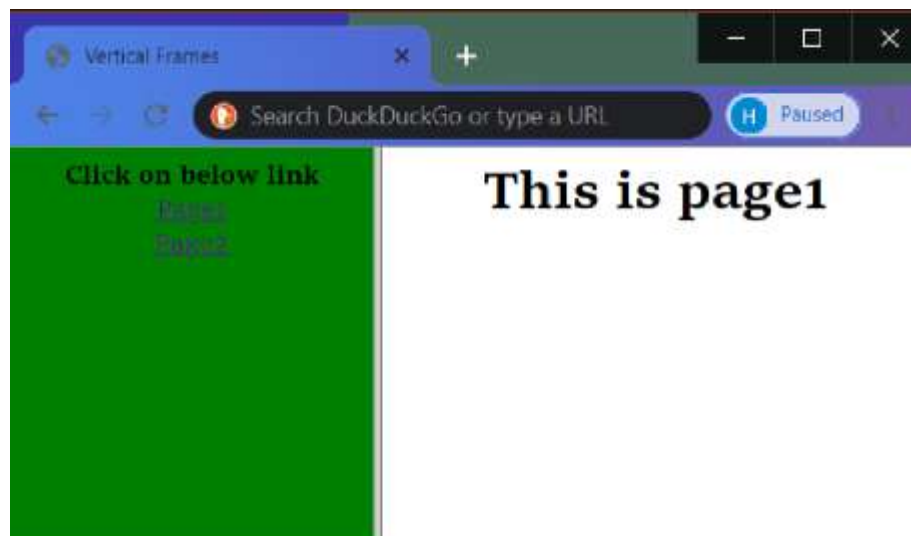
menu.html

```
<html>
<head>
    <title>Menu</title>
</head>
<body bgcolor="green">
    <center><b>Click on below link</b>
        <br><a href="page1.html" target="display">Page1
        <br><a href="page2.html" target="display">Page2
    </center>
</body>
</html>
```

Defual.html

```
<html>
<head>
    <title>Vertical Frames</title>
</head>
<frameset cols="40%,60%">
    <frame src=menu.html>
    <frame src=default.html name="display">
</frameset>
</html>
```

Output



Cascading Style Sheets:

Style sheets represent the World Wide Web consortium's effort to improve on the tag and attribute based style of formatting. Style sheets provide a way of customizing whole pages all at once and in much richer detail than the simple use of tags and attributes. The format of style sheet will be:

```
<style type="text/css">
    selector{property:value;property:value;}
    selector{property:value;property:value;}
</style>
```

Every line in <style> tag is called as a „**Rule**“ and a style rule has two parts:

- a. Selector.
- b. Set of declarations.

A selector is used to create a link between the rule and the HTML tag. The declaration has two parts again:

- a. Property.
- b. Value.

A property specifies additional information and value specifies property value. For example:

```
<style type="text/css">
body {background-color: #d0e4fe;}
h1 {
color: orange;
text-align: center;
}
p {
font-family: "Times New Roman";
font-size: 20px;
}
</style>
```

If we add above code in the <head> element of web page, entire web page will be displayed in

various styles given in style element.

Style sheets are implemented with cascading style sheets specification. Conventionally styles are cascaded i.e., we don't have to use just a single set of styles inside a document, but we can import as many styles as we like. There are three mechanisms by which we can apply styles to our HTML documents:

1. Inline Style sheets.
2. Embedded Style sheets.
3. External Style sheets.

Inline Style Sheets:

Inline style sheets mix content with presentation. To use inline styles we use style attribute in the relevant tag.

Example:

```
<html>
  <head>
    <title>HTML Tables</title>
  </head>
  <body bgcolor="pink">
    <center>
      <h1>Creating HTML Tables</h1><br>
      <table border="2" cellpadding="4" cellspacing="4">
        <tr>
          <th colspan="2" style="background-color:red"> WebSites</th>
        </tr>
        <tr>
          <th style="background-color:blue">MailSites</th>
          <th style="background-color:green">JobSites</th>
        </tr>
        <tr>
          <td style="background-color:grey">Gmail</td>
          <td style="background-color:aqua">Naukri</td>
```

```
</tr>
<tr>
    <td style="background-color:yellow">Yahoo</td>
    <td style="background-color:purple">JobStreet</td>
</tr>
</table>
</center>
</body>
</html>
```

Output:



Embedded Style sheets:

An embedded style sheet is used when a single document has a unique style. We define internal styles in the head section of a HTML page by using „<style>“ tag. The styles defined using embedded style sheets are applied throughout the page and we put the styles into oneplace.

Example:

```
<html>
  <head>
    <title>Embedded Style sheets</title>
    <style type="text/css">
      body{background-color:
        pink;}
      h1 {
        color:orang
        e;
        text-align:
        center;
        }
      p {
        font-family:  "Times  New
        Roman";
        font-size: 20px;
        }
    </style>
  </head>
  <body>
    <h1>Embedded Style Sheets</h1><br>
    <p>This is a paragraph
  </body>
</html>
```

Output:



External Style Sheets:

External style sheets are just that the style sheets are stored separately from our web page. These are useful especially if we are setting the styles for an entire website. When we change the styles in external style sheet we change the styles of all pages. We use „<link>“ element to access the style sheet file defined into our web page. The format of <link> element is:

```
<link rel="stylesheet" type="text/css" href="extstylesheet.css">
```

Example:

extern.css:

```
body {background-color: #d0e4fe;}  
h1 {  
color: orange; text-align: center;  
}  
p {  
font-family: "Times New Roman"; font-size: 20px;  
}
```

extern.html:

```
<html>  
<head>  
<title>External Style Sheets</title>
```

```
<link rel="stylesheet" type="text/css" href="extern.css">
</head>
<body>
<h1>External Style Sheets</h1><br>
<p>This is a paragraph
</body>
</html>
```

Output:

